found by Hjalmar for the K rays of the lighter elements, and by Auger and Dauvillier* for the L rays of some of the heavier elements, and may arise from electron transitions in atoms which have lost more than one electron. The line 1380·6 X.U. lies a little way from the graph of the line called "*t*" by these last-named workers.

This work was carried out while one of us was a Fred Knight and University Research Scholar. In conclusion, we wish to express our gratitude to Prof. T. H. Laby and to Mr. J. S. Rogers for their helpful interest and advice during the progress of the investigation, and to the latter for his assistance in writing this report.

The Ratios of the Specific Heats of Nitrogen at Atmospheric Pressure and at Temperatures between 10° *C. and* −183° *C.*

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These measurements of the ratios of the specific heats of nitrogen, at room and at lower temperatures, have been made, and the results have been calculated, in exactly the same way as those obtained when air and hydrogen were used. The latter were described in a former paper, (1) in which will be found a description of the apparatus and an account of the method of dealing with the experimental results. In the work which follows the same apparatus was employed, the only changes being that a new platinum wire (from the same reel) was used on the thermometer, and, owing to breakage, a rather finer quartz suspension had been put on the galvanometer. However, as the apparatus had to be dismantled for transference to another laboratory I thought it desirable to confirm some of the effects and relationships previously noted; therefore, a full set of experiments with the four expansion vessels was made at room temperature. It may be recalled that the method has been especially worked out, using small expansion vessels in order that values of the ratios of the specific heats can be readily obtained at different temperatures. Up to the present time work has been restricted to determinations at temperatures below that of the laboratory. Several vessels are used in order to allow a correction to be made for a systematic change in the apparent value of the ratio, which depends on the size of the vessel employed.

* 'Comptes Rendus,' vol. 176, p. 1297 (1923).

The nitrogen was supplied by the British Oxygen Company, and was stated to contain less than ½ of 1 per cent. of impurity, mostly oxygen. I am indebted to Mr. D. Newitt for six analyses of gas taken from the apparatus at various times during the course of the experiments. All of these show that the percentage impurity was less than the amount stated above.

The observations at $-78°$ C., reduced in the way previously described, are given on p. 126. They show the steadiness of the experimental conditions and the accuracy with which these could be reproduced. The quantity γ_a, tabulated in the last column, is the apparent value of the ratio of the specific heats calculated from the well-known relation

$$s_p/s_v = \frac{\log p_1/p_2}{\log p_1/p_2 - \log \theta_1/\theta_2}, \tag{A}$$

where p_1 and p_2 are the initial and final pressures in centimetres of mercury, and θ_1 and θ_2 the initial and final temperatures on the absolute scale. The values of γ_a thus obtained vary systematically with the size of the vessel used. By extrapolating linearly, a procedure which has been shown to be extraordinarily accurate over a considerable range in the size of the vessels employed, and which is again confirmed, a value indicated by γ' is deduced. This is the apparent value of the ratio which would have been obtained if the measurements had been made using a vessel of infinite volume. This extrapolated value still requires two corrections, (*a*) the so-called radiation correction, and (*b*) a theoretical correction.

Fundamental Interval, Freezing Point, and Heating Effect Observations in Centimetres of Bridge Wire.

Date, 1925.	Temp. °C.	1 cell.	2 cells.	
14·10	0	705·72	706·11	F.P. 705·59
	99·985	973·43	973·72	F.I. 267·78
9·11	0	705·78	706·17	F.P. 705·65
	99·551	972·33	972·61	F.I. 267·79

Heating Effects (subtractive).

Date, 1925.	Temp. °C.	1 cell.	2 cells.
14·10	0	0·13	0·52
	100	0·10	0·39
20·10	11	0·13	0·51
29·10	— 78	0·16	0·63
4·11	—183	0·17	0·68
9·11	0	0·13	0·52
	100	0·09	0·37

In the two experiments made in 1924, with the apparatus as used in the air and hydrogen experiments, the constants of the thermometer were F.P. = 813·88 and F.I. = 310·95. The heating effects being 0·58 at 11° C., and 0·21 at − 183° C.

Reduced Observations.

Columns (*a*) and (*d*) give the values of θ_1 and θ_2 expressed in centimetres of bridge wire, after the application of all corrections.

Column (*b*) gives the bridge wire setting for the observation of θ_2 and (*c*) the resulting kick deflection.

October 29, 1925. 300 c.c. bulb.

p_2	p_1	*a*	*b*	*c*	*d*	θ_2	θ_1	γ_a
75·65	80·590	489·72	20·4	35	480·08	191·321	194·763	1·3928
	562	72	2	1	09	325	763	3942
	538	72	4	29	12	335	763	3927
	518	73	3	2	18	356	766	3981
	496	75	4	8	24	377	773	3953
	465	76	4	0	29	395	777	3964
	442	76	5	8	34	413	777	3961
	416	76	5	−1	40	435	777	3953
	392	75	7	26	44	449	773	3950
	369	75	6	2	48	464	773	3951
	344	75	7	2	58	499	773	3952
	322	77	9	29	62	513	780	3933
	299	79	9	13	71	545	788	3921
	273	82	9	8	74	555	799	3949

October 29, 1925. 35 c.c. bulb.

p_2	p_1	*a*	*b*	*c*	*d*	θ_2	θ_1	γ_a
75·59	80·528	489·78	20·6	17	480·29	191·396	194·786	1·3840
	516	78	8	38	24	378	786	3880
	508	78	6	11	36	421	786	3821
	494	78	5	2	37	425	786	3830
	487	78	6	10	37	425	786	3837
	477	79	5	1½	37	425	789	3852
	466	79	6	9	38	428	789	3859
	454	79	5	−4	44	449	789	3838
	440	79	6	1	48	463	789	3830
	430	79	8	20	46	456	789	3851
	418	79	6	1	48	463	789	3853
	410	78	7	6½	51	474	786	3840
	397	78	7	4	54	485	786	3836

Other values of γ_a obtained under similar conditions: 1·3850, 1·3834, 1·3822, 1·3834, 1·3836, 1·3837, 1·3842, 1·3848.

Summarised Results for Nitrogen.

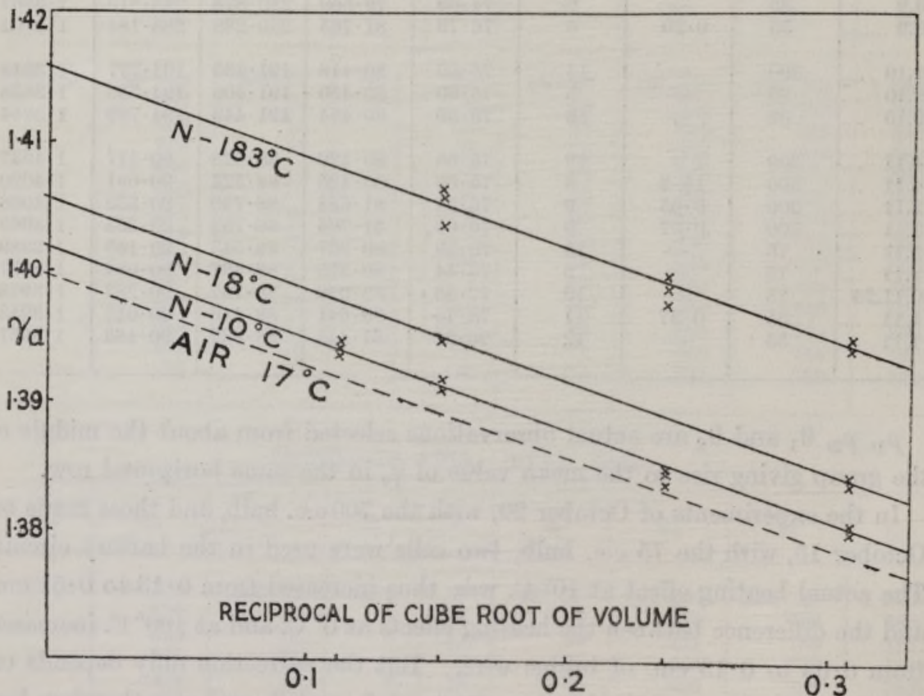
Date, 1925.	Vol., c.c.	Time, secs.	No. of Obs.	p_2, cms.	p_1, cms.	θ_2, Abs.	θ_1, Abs.	γ_a, Means.
15.10	700	0·51	9	76·17	80·975	277·618	282·467	1·3936
16.10	700	0·60	10	76·15	80·916	277·378	282·186	1·3945
16.10	700	1	9	76·17	80·945	277·443	282·242	1·3931
19.10	700	—	8	75·95	80·750	280·000	284·878	1·3935
19.10	700	1	9	75·94	80·743	279·923	284·822	1·3933
20.10	700	—	10	75·19	80·055	278·881	283·862	1·3942
9.11	700	$<0·5$	7	75·73	80·592	277·538	282·467	1·3945
22.10	300	0·33	10	73·84	78·715	284·132	289·288	1·3918
25.10	300	0·45	9	75·05	79·950	279·285	284·298	1·3911
14.10	75	0·31	10	75·98	80·864	279·273	284·150	1·3851
15.10	75	0·21	10	76·18	81·123	277·229	282·119	1·3840
18.11.24	75	—	10	77·35	82·283	279·450	284·291	1·3849
23.9	35	—	6	74·69	79·560	280·878	285·815	1·3807
28.9	35	0·20	8	76·79	81·765	280·288	285·184	1·3802
29.10	300	—	14	75·65	80·416	191·435	194·777	1·3948
29.10	35	—	8	75·60	80·450	191·408	194·736	1·3838
29.10	35	—	13	75·59	80·454	191·449	194·789	1·3844
4.11	300	1	12	75·65	80·489	88·528	90·117	1·4037
4.11	300	1½–2	8	75·68	80·425	88·524	90·081	1·4020
12.11	300	0·65	9	76·94	81·688	88·785	90·330	1·4058
12.11	300	0·57	9	76·90	81·795	88·758	90·354	1·4065
5.11	75	—	13	75·39	80·257	88·548	90·143	1·3989
11.11	75	—	9	76·44	80·375	88·497	90·084	1·3997
20.11.24	75	—	10	77·36	82·079	88·767	90·283	1·3978
4.11	35	0·37	11	75·70	80·641	88·416	90·012	1·3943
11.11	35	—	12	76·62	81·445	88·637	90·183	1·3951

p_1, p_2, θ_1 and θ_2 are actual observations selected from about the middle of the group giving rise to the mean value of γ_a in the same horizontal row.

In the experiments of October 20, with the 700 c.c. bulb, and those made on October 15, with the 75 c.c. bulb, two cells were used in the battery circuit. The actual heating effect at 10° C. was thus increased from 0·13 to 0·51 cm. and the difference between the heating effects at 0° C. and at 100° C. increased from 0·03 to 0·13 cm. of bridge wire. But the correction only depends on the difference between the heating effects at θ_1 and θ_2 and was therefore less than 0·01 cm. of bridge wire. That is, the additive correction to γ_a was less than one part in 3000, and in all the other experiments on nitrogen it was negligible. As the heating effects were constantly re-determined under the *actual* experimental conditions, and very consistent values were obtained, the correction could always be made with great accuracy, even when it was considerably larger, as it was in my experiments on hydrogen at room temperature.

Actual time measurements were not made in every case as the chronograph was required for other work. The large values of the time given in four instances were estimated by direct observation. The consistency of the values of γ_a obtained with the largest bulb, when the times were varied from less than 0·5 to about 1 second, shows that overshooting does not affect the results. Similar evidence was given in my paper on air and hydrogen. Probably rather too much cooling has taken place during the interval between the commencement of the expansion and the time at which the galvanometer circuit is made, when this interval is about 1 second in the cases of (*a*) the 700 c.c. bulb at room temperature and (*b*) the 300 c.c. bulb at $-$ 183° C.

In the figure the mean results obtained with each bulb are plotted against



the reciprocal of a linear dimension of the vessel employed, *i.e.*, the cube root of the volume. For the bulbs employed the values of these abscissæ are 0·1128, 0·1516, 0·2371 and 0·3072 respectively. The linear relation previously shown to exist is amply verified. The points given by the intersections of these lines with that of infinite volume give values of γ' at the three temperatures at which experiments were made.

These values are, in the case of nitrogen, 1·4019 at 10° C., 1·4056 at $-$ 78°C., and 1·4158 at − 183° C. The dotted line shows results previously obtained with air and is included for comparison.

The values of γ' require two further corrections (a) the so-called radiation correction; (b) the theoretical correction.

(a) The apparent values of γ measured first with a bright and then with a blackened wire, give rise to an additive correction. There seems to be no doubt as to the magnitude of this correction in the case of air at room temperature, as several observers have obtained closely agreeing values. The value taken is 0·0021. Lummer and Pringsheim (2) added this amount to their experimental values for air, hydrogen, etc., and Partington and Howe (3) apply the correction in the same way. I have given reasons for assuming that the correction is not a pure radiation effect, and it should vary in the inverse ratio of the thermal conductivity of the gas surrounding the thermometer wire. For hydrogen the correction is then lowered to 0·0004 (1).

Since the publication of my paper, Mandell (4) has shown that fine wire thermocouples record temperatures which depend, amongst other factors, on the nature of the surrounding gas. The ratio of the differences found in air and in hydrogen is about 5; as I had assumed.

The radiation correction in the case of nitrogen is the same as for air, *i.e.*, 0·0021 at room temperature. The values at lower temperatures have been calculated by the application of Stefan's law.

(b) Equation A (p. 125) applies only to perfect gases which obey the law

Summary.

Final Values for the Ratios of the Specific Heats of Nitrogen.

Temp. °C.	γ'.	Radiation correction r.	$\gamma'_1 = \gamma' + r$.	Thermodynamical correction factor $(1+f)$.	Final values of γ.	Values obtained by other Observers.
10	1·4019	0·0021	1·4040	1·0010 (B) (C)	1·4054	1·4045 P & H (20°C.) 1·406 D, G & P (0°C.)
− 78	1·4056	0·0005	1·4061	1·0030 (B) (C)	1·4103	
− 183	1·4158	—	1·4158	1·0310 (B) 1·0208 (C)	1·4596 1·4454	1·447 V (−190° C.).

P & H: reference 3.
D, G & P: „ 8.
V: „ 9.

$p\,v = R\theta$. By assuming one of the many characteristic gas equations, it can be shown that γ' must be multiplied by a factor $(1 + f)$ in order to correct for the difference between the properties of the gas in question and those of a perfect gas.

I have given the corrections calculated by using the gas equations proposed by Callendar (C) (5) and Berthelot (B) (6), as the difference in the magnitudes of these corrections is large at very low temperatures. The constants used in calculating the Callendar correction are those given by Prof. Callendar in his paper on the 'Thermodynamical Correction of the Gas Thermometer.' For nitrogen the critical pressure is 33 atmos., and the critical temperature 126·5° Abs.

Accuracy of the Measurements.

The effect of observational errors on the values of γ_a was considered in my previous paper. In all the experiments by this method, as is shown by illustrative tables of reduced observations both in this and in my earlier paper, the maximum deviation of an individual from the mean result obtained with any one vessel, at any one temperature, is of the order 0·002 to 0·003. These differences are not greater than those found in the results of all other observers.

Accuracy of the Extrapolation.

Previous evidence of the parallelism of the linear relationships between γ_a and the reciprocal cube root of the volume of the vessel is confirmed by these experiments. As it is shown that this linear relation holds over a considerable range in the size of the vessel used, the accuracy of the extrapolated value γ' will be of the same order as that of the individual means.*

Recent Work by Other Observers.

References to earlier measurements of the ratio of the specific heats of nitrogen will be found in the appended list of papers. Partington and Howe measured the value of γ for nitrogen, using a 60-litre vessel and a compensated thermometer of platinum wire 0·01 mm. in diameter. An Einthoven string galvanometer was employed. They found it necessary to make a somewhat

* The proper interpretation of my results for hydrogen at room temperature, which is in accordance with all the experimental evidence, indicates that my value of γ' for hydrogen is correct to within 0·002, and not only within 0·012 as is suggested by Partington and Howe (7).

provisional, but relatively large, correction of about 1/2 per cent. in order to correct for the so-called lag of their compensated thermometer. The data for this correction were obtained from measurements made with uncompensated wires surrounded by air only. I have previously pointed out that the application of the full correction calculated from these data is not justifiable.*

The excellent series of measurements by Dixon and his collaborators (8) at temperatures above that of the room and Valentiner's (9) measurement at about 82·5° Abs. are especially valuable. Dixon, Campbell and Parker made direct measurements of the velocity of sound in nitrogen. My measurement at room temperature is in excellent agreement with that deduced from their observations. Valentiner used the Kundt's tube method, and obtained the value 1·447 at about − 190° C.

The values of the molecular specific heats at constant pressure and at constant volume can be calculated from the final values of those given above. The specific heat values given in the following table have been calculated by using (B) Berthelot's and (C) Callendar's gas equations. The gas constant has been taken as 1·985, and the molecular weight of nitrogen as 28·02.

NITROGEN.

Values of the Specfic Heats.

θ° A.	$\gamma - 1$.	$S_p - S_v$.	S_v.	s_p.	Direct measurements of s_p.
283	0·4054	1·995	4·922	0·2468	0·2492 S & H, 0·242*
195	0·4103	2·014	4·910	0·2471	
90	0·4596 (B)	2·292 (B)	4·988 (B)	0·2597	0·2556 S & H, 0·250*
	0·4454 (C)	2·189 (C)	4·914 (C)	0·2535	

* By linear extrapolation of Scheel and Heuse's experimental results, reference 10.

The only direct observation of the specific heat of nitrogen at constant pressure and at the temperature of liquid air is that of Scheel and Heuse (10). The value of the ratio deduced from their measurements, using Berthelot's equation,

* The applicability of the full correction to their experiments on hydrogen is still more doubtful as any thermometric effect giving rise to such a correction would depend on the nature of the gas surrounding the thermometer wire. Therefore it seems probable that Partington and Howe's value of γ for hydrogen, which, when corrected for radiation by the method I have adopted, would be 1·4096, is somewhat too high. My own value is 1·4070.

K 2

is $1 \cdot 468$. Scheel and Heuse point out that when their results are calculated, using the gas equation employed by Valentiner, their value of γ is reduced to $1 \cdot 459$, which is about 1 per cent. in excess of Valentiner's estimation of this quantity.

The characteristic equation used by Valentiner was one deduced by Bestelmeyer and himself (11), as the result of an experimental study of the behaviour of nitrogen at liquid-air temperatures.

It is interesting to note the excellent agreement between Valentiner's value for γ and my own, when the latter is calculated by the aid of Callendar's equation. As the Bestelmeyer and Valentiner relation represents the actual properties of nitrogen over a small range of temperature in the neighbourhood of -190° C., this agreement shows that Callendar's type of equation, in addition to its precision at higher temperatures, certainly represents the properties of nitrogen near its saturation-point with greater exactitude than does that proposed by Berthelot.

I have previously pointed out that the extrapolations in Scheel and Heuse's experiments may be somewhat doubtful (1). This is more especially the case in their experiments on nitrogen, as flows of two widely differing values were used only.

My value for the specific heat of nitrogen at constant pressure and at -183° C. is within $0 \cdot 3$ per cent. of the mean of the value they deduce and that obtained, viz., $0 \cdot 250$, by a linear extrapolation of their results.

Variation of Molecular Specific Heat with Temperature.

Dixon (8*b*) gives $S_v = 4 \cdot 922 + 0 \cdot 00041\, t$, as representing the results of the experiments on nitrogen at atmospheric pressure and between 0° and 1000° C. At high temperatures $S_p - S_v = 1 \cdot 985$, so that $S_p = 6 \cdot 91 + 0 \cdot 00041\, t$. Callendar (5) has shown that his equation gives the relation $S_p = S_o + an(n+1)\, cp/\theta$, where S_o is the limiting value of S_p at zero pressure and temperature θ.

Callendar assumes that in this ideally rarefied condition the gas is perfect and S_o is independent of the temperature. If the equation representing Dixon's experimental results is combined with Callendar's theoretical expression, in which the term involving $p/\theta^{2 \cdot 5}$ is negligible at high temperatures, we may imagine Dixon's linear variation of S_p to hold for the zero-pressure condition, and $6 \cdot 91$ becomes the value of S_o at 0° C. The molecular specific heats may then be calculated over the complete range of temperatures and pressures.

The agreement with my experimental measurements at atmospheric pressure is very good, as is shown in the adjoining table :—

Temp. $t°$ C.	S_p (exp.).	S_p (calc.).
10	6·92	6·92
− 78	6·92	6·91
− 183	7·10	7·07

Note.—In referring to a criticism in my former paper, Partington and Howe (7) say, " Brinkworth is therefore in error in stating that an emergent stem correction was applied " in these experiments. My statement referred explicitly (pp. 539–540) to the value, 1·4001 (air), obtained by Partington (12). In an example which Partington gives in detail, and which constitutes one of the individual experiments leading to this value, the emergent stem correction is given as 0·01° C. The paragraph from which they quote contained no reference to experiments made by them conjointly.

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